Midterm Group Project - Part 1: Operational Databases, Data Warehousing, and ETL

Part 1: Environment Setup

Task 1.1: Start the Docker Compose Environment

```
[+] up 10/10
 ✔Image part01-etl-service      Built                                                                    22.4s
 ✔Network part01_week03_network Created                                                                  0.0s
 ✔Volume part01_sales_data      Created                                                                  0.0s
 ✔Volume part01_hr_data         Created                                                                  0.0s
 ✔Volume part01_warehouse_data  Created                                                                  0.0s
 ✔Container week03_sales_db     Healthy                                                                  10.8s
 ✔Container week03_hr_db        Healthy                                                                  11.3s
 ✔Container week03_warehouse_db Healthy                                                                  11.3s
 ✔Container week03_pgadmin      Created                                                                  0.1s
 ✔Container week03_etl          Created                                                                  0.1s
PS C:\Users\ksogodogo\ism6562s26-class\midterm\part01> docker compose ps
NAME                  IMAGE                 COMMAND                  SERVICE        CREATED              STATUS                        P
ORTS
week03_hr_db          postgres:16           "docker-entrypoint.s…"   hr-db          About a minute ago   Up About a minute (healthy)   0
.0.0.0:5502->5432/tcp, [::]:5502->5432/tcp
week03_pgadmin        dpage/pgadmin4:8.13   "/entrypoint.sh"         pgadmin        About a minute ago   Up 57 seconds                 0
.0.0.0:8200->80/tcp, [::]:8200->80/tcp
week03_sales_db       postgres:16           "docker-entrypoint.s…"   sales-db       About a minute ago   Up About a minute (healthy)   0
.0.0.0:5501->5432/tcp, [::]:5501->5432/tcp
week03_warehouse_db   postgres:16           "docker-entrypoint.s…"   warehouse-db   About a minute ago   Up About a minute (healthy)   0
.0.0.0:5503->5432/tcp, [::]:5503->5432/tcp
```

Task 1.2: Connect to Each Database via psql

Connect to the sales database and list tables:

```
PS C:\Users\ksogodogo\ism6562s26-class\midterm\part01> docker compose exec sales-db psql -U student -d sales -c '\dt'
          List of relations
 Schema |   Name    | Type  |  Owner
--------+-----------+-------+---------
 public | customers | table | student
 public | orders    | table | student
 public | products  | table | student
(3 rows)
```

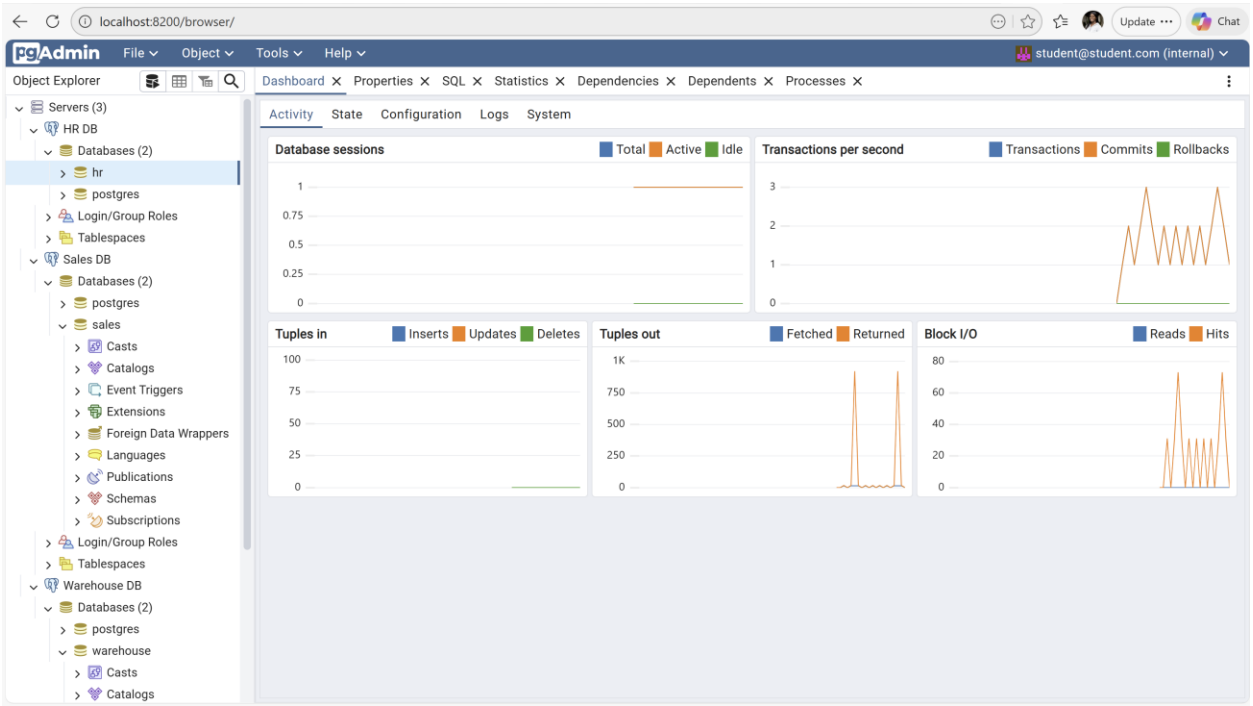
Connect to the HR database and list tables:

```
PS C:\Users\ksogodogo\ism6562s26-class\midterm\part01> docker compose exec hr-db psql -U student -d hr -c '\dt'
           List of relations
 Schema |    Name     | Type  |  Owner
--------+-------------+-------+---------
 public | departments | table | student
 public | employees   | table | student
 public | timesheets  | table | student
(3 rows)
```

Connect to the warehouse database and list tables:

```
PS C:\Users\ksogodogo\ism6562s26-class\midterm\part01> docker compose exec warehouse-db psql -U student -d warehouse -c '\dt'
            List of relations
 Schema |     Name     | Type  |  Owner
--------+--------------+-------+---------
 public | dim_customer | table | student
 public | dim_date     | table | student
 public | dim_employee | table | student
 public | dim_product  | table | student
 public | fact_sales   | table | student
(5 rows)
```

Task 1.3: Register All Three Servers in pgAdmin



Part 2: Explore Operational Databases

Task 2.1: Examine the Sales Schema

Connect to the Sales database:

```
PS C:\Users\ksogodogo\ism6562s26-class\midterm\part01> docker compose exec sales-db psql -U student -d sales
psql (16.12 (Debian 16.12-1.pgdg13+1))
Type "help" for help.

sales=#
```

Inspect sales database table structures:

Customers Table

```
sales=# \d customers
                                        Table "public.customers"
   Column    |            Type             | Collation | Nullable |                   Default
-------------+-----------------------------+-----------+----------+-----------------------------------------------
 customer_id | integer                     |           | not null | nextval('customers_customer_id_seq'::regclass)
 first_name  | character varying(100)      |           | not null |
 last_name   | character varying(100)      |           | not null |
 email       | character varying(255)      |           | not null |
 city        | character varying(100)      |           |          |
 state       | character varying(2)        |           |          |
 created_at  | timestamp without time zone |           |          | now()
Indexes:
    "customers_pkey" PRIMARY KEY, btree (customer_id)
    "customers_email_key" UNIQUE CONSTRAINT, btree (email)
    "idx_customers_email" btree (email)
    "idx_customers_state" btree (state)
Referenced by:
    TABLE "orders" CONSTRAINT "orders_customer_id_fkey" FOREIGN KEY (customer_id) REFERENCES customers(customer_id)
```

Products Table

```
sales=# \d products
                                          Table "public.products"
    Column    |            Type             | Collation | Nullable |                   Default
--------------+-----------------------------+-----------+----------+-----------------------------------------------
 product_id   | integer                     |           | not null | nextval('products_product_id_seq'::regclass)
 product_name | character varying(200)      |           | not null |
 category     | character varying(100)      |           | not null |
 unit_price   | numeric(10,2)               |           | not null |
 created_at   | timestamp without time zone |           |          | now()
Indexes:
    "products_pkey" PRIMARY KEY, btree (product_id)
    "idx_products_category" btree (category)
Referenced by:
    TABLE "orders" CONSTRAINT "orders_product_id_fkey" FOREIGN KEY (product_id) REFERENCES products(product_id)
```

Orders Table

```
sales=# \d orders
                                  Table "public.orders"
   Column    |     Type      | Collation | Nullable |                 Default
-------------+---------------+-----------+----------+-----------------------------------------
 order_id    | integer       |           | not null | nextval('orders_order_id_seq'::regclass)
 customer_id | integer       |           | not null |
 product_id  | integer       |           | not null |
 quantity    | integer       |           | not null |
 total_price | numeric(10,2) |           | not null |
 order_date  | date          |           | not null |
Indexes:
    "orders_pkey" PRIMARY KEY, btree (order_id)
    "idx_orders_customer_id" btree (customer_id)
    "idx_orders_order_date" btree (order_date)
    "idx_orders_product_id" btree (product_id)
Check constraints:
    "orders_quantity_check" CHECK (quantity > 0)
Foreign-key constraints:
    "orders_customer_id_fkey" FOREIGN KEY (customer_id) REFERENCES customers(customer_id)
    "orders_product_id_fkey" FOREIGN KEY (product_id) REFERENCES products(product_id)
```

In the sales database, each table, Customers, Products, and Orders, has a column with a unique identifier as the primary key, which are respectively customer_id, product_id, and order_id, and they are generated automatically and identify every row. Additionally, this schema puts indexes on columns that are most likely to appear in queries, such as foreign keys, order date, email, etc. Some indexes also act as rules, like the one on email that both speeds up searches and makes sure no two customers share the same email.

Here, normalization avoids redundancy in multiple ways. For instance, the Orders table uses foreign keys to connect to the other two tables. Customer information is only stored in the Customers table, so that every order only has a customer ID instead of repeating a customer's name and address on every order. Similarly, for Products, the order only has the product's ID, not all details about the product like its name or price. This ensures that everything is stored only once and avoids duplicate data. Customers' information is in Customers, product details are in Products and orders use IDs to refer to each item in those tables.

Task 2.2: Examine the HR Schema

Connect to the HR database

```
sales=# exit
PS C:\Users\ksogodogo\ism6562s26-class\midterm\part01> docker compose exec hr-db psql -U student -d hr
psql (16.12 (Debian 16.12-1.pgdg13+1))
Type "help" for help.
```

Inspect HR database table structures:

Departments Table

```
hr=# \d departments
                                          Table "public.departments"
     Column      |          Type          | Collation | Nullable |                     Default
-----------------+------------------------+-----------+----------+-----------------------------------------------------
 department_id   | integer                |           | not null | nextval('departments_department_id_seq'::regclass)
 department_name | character varying(100) |           | not null |
 location        | character varying(100) |           | not null |
Indexes:
    "departments_pkey" PRIMARY KEY, btree (department_id)
    "departments_department_name_key" UNIQUE CONSTRAINT, btree (department_name)
Referenced by:
    TABLE "employees" CONSTRAINT "employees_department_id_fkey" FOREIGN KEY (department_id) REFERENCES departments(department_id)
```

Employee Table

```
hr=# \d employees
                                          Table "public.employees"
    Column     |          Type          | Collation | Nullable |                     Default
---------------+------------------------+-----------+----------+-----------------------------------------------
 employee_id   | integer                |           | not null | nextval('employees_employee_id_seq'::regclass)
 first_name    | character varying(100) |           | not null |
 last_name     | character varying(100) |           | not null |
 email         | character varying(255) |           | not null |
 department_id | integer                |           | not null |
 hire_date     | date                   |           | not null |
 job_title     | character varying(100) |           | not null |
Indexes:
    "employees_pkey" PRIMARY KEY, btree (employee_id)
    "employees_email_key" UNIQUE CONSTRAINT, btree (email)
    "idx_employees_department_id" btree (department_id)
    "idx_employees_hire_date" btree (hire_date)
Foreign-key constraints:
    "employees_department_id_fkey" FOREIGN KEY (department_id) REFERENCES departments(department_id)
Referenced by:
    TABLE "timesheets" CONSTRAINT "timesheets_employee_id_fkey" FOREIGN KEY (employee_id) REFERENCES employees(employee_id)
```

Timesheet Table

```
hr=# \d timesheets
                                        Table "public.timesheets"
    Column     |     Type     | Collation | Nullable |                     Default
---------------+--------------+-----------+----------+-----------------------------------------------
 timesheet_id | integer      |           | not null | nextval('timesheets_timesheet_id_seq'::regclass)
 employee_id  | integer      |           | not null |
 work_date    | date         |           | not null |
 hours_worked | numeric(4,2) |           | not null |
Indexes:
    "timesheets_pkey" PRIMARY KEY, btree (timesheet_id)
    "idx_timesheets_employee_id" btree (employee_id)
    "idx_timesheets_work_date" btree (work_date)
    "timesheets_employee_id_work_date_key" UNIQUE CONSTRAINT, btree (employee_id, work_date)
Check constraints:
    "timesheets_hours_worked_check" CHECK (hours_worked >= 0::numeric AND hours_worked <= 24::numeric)
Foreign-key constraints:
    "timesheets_employee_id_fkey" FOREIGN KEY (employee_id) REFERENCES employees(employee_id)
```

The HR database also has three tables: Departments, Employees, and Timesheets; each have a primary key, which are department_id, employee_id, and timesheet_id respectively, to uniquely identify every row. Here, the schema also places indexes on columns that frequently appear in queries and some of these indexes serve prevents duplicate department entries and there is a combined constraint on employee_id and work_date that stops the same employee from logging hours multiple times on the same day.

Just like in the sale database, everything stored in one place; department details are in Departments table, employee details are in Employee table and Timesheets uses IDs to refer back to both tables. More explicitly, instead of keeping the department's name and location, Employees table references a department_id and similarly, the Timesheets table only holds

an employee_id instead of duplicating an employee's name or department on every single entry.

Task 2.3: Run Queries Against the Operational Database

Sales Database Queries:

-- Query 1: Total revenue by product category

```
sales=# SELECT p.category,
       COUNT(*) AS order_count,
       SUM(o.total_price) AS total_revenue
FROM orders o
JOIN products p ON o.product_id = p.product_id
GROUP BY p.category
ORDER BY total_revenue DESC;
  category   | order_count | total_revenue
-------------+-------------+---------------
 Electronics |          15 |       7199.81
 Furniture   |           7 |       2789.92
 Books       |           5 |        395.91
(3 rows)
```

-- Query 2: Customers with the most orders

```
sales=# SELECT c.first_name, c.last_name, c.state,
       COUNT(*) AS order_count
FROM orders o
JOIN customers c ON o.customer_id = c.customer_id
GROUP BY c.customer_id, c.first_name, c.last_name, c.state
ORDER BY order_count DESC;
 first_name | last_name | state | order_count
------------+-----------+-------+-------------
 Alice      | Johnson   | FL    |           5
 Carol      | Williams  | GA    |           4
 Bob        | Smith     | FL    |           4
 Eva        | Davis     | NC    |           3
 David      | Brown     | FL    |           3
 Frank      | Miller    | FL    |           3
 Grace      | Wilson    | TN    |           3
 Henry      | Taylor    | GA    |           2
(8 rows)
```

HR Database Queries:

-- Query 3: Employee count per department

```
hr=# SELECT d.department_name, d.location,
       COUNT(*) AS employee_count
FROM employees e
JOIN departments d ON e.department_id = d.department_id
GROUP BY d.department_id, d.department_name, d.location
ORDER BY employee_count DESC;
 department_name | location | employee_count
-----------------+----------+----------------
 Human Resources | Tampa    |              3
 Marketing       | Orlando  |              3
 Sales           | Miami    |              3
 Engineering     | Tampa    |              3
(4 rows)
```

-- Query 4: Total hours worked per department (Jan 2026)

```
hr=# SELECT d.department_name,
       SUM(t.hours_worked) AS total_hours,
       ROUND(AVG(t.hours_worked), 1) AS avg_hours_per_entry
FROM timesheets t
JOIN employees e ON t.employee_id = e.employee_id
JOIN departments d ON e.department_id = d.department_id
WHERE t.work_date BETWEEN '2026-01-01' AND '2026-01-31'
GROUP BY d.department_id, d.department_name
ORDER BY total_hours DESC;
 department_name | total_hours | avg_hours_per_entry
-----------------+-------------+---------------------
 Engineering     |      158.00 |                 7.9
 Sales           |      139.50 |                 7.8
 Marketing       |      133.00 |                 7.8
 Human Resources |      131.00 |                 7.7
(4 rows)
```

Part 3: Data Warehouse Concepts

Task 3.1: Examine the Star Schema

Connect to the warehouse database and inspect all table structures:

\d dim_date

```
warehouse=# \d dim_date
                        Table "public.dim_date"
    Column    |         Type          | Collation | Nullable | Default
--------------+-----------------------+-----------+----------+---------
 date_key     | integer               |           | not null |
 full_date    | date                  |           | not null |
 year         | integer               |           | not null |
 quarter      | integer               |           | not null |
 month        | integer               |           | not null |
 month_name   | character varying(20) |           | not null |
 day_of_month | integer               |           | not null |
 day_of_week  | integer               |           | not null |
 day_name     | character varying(20) |           | not null |
 is_weekend   | boolean               |           | not null |
Indexes:
    "dim_date_pkey" PRIMARY KEY, btree (date_key)
    "dim_date_full_date_key" UNIQUE CONSTRAINT, btree (full_date)
Referenced by:
    TABLE "fact_sales" CONSTRAINT "fact_sales_date_key_fkey" FOREIGN KEY (date_key) REFERENCES dim_date(date_key)
```

```
    "dim_customer_source_customer_id_key" UNIQUE CONSTRAINT, btree (source_customer_id)
Referenced by:
    TABLE "fact_sales" CONSTRAINT "fact_sales_customer_key_fkey" FOREIGN KEY (customer_key) REFERENCES dim_customer(customer_key)
```

\d dim_product

```
warehouse=# \d dim_product
                                        Table "public.dim_product"
      Column       |          Type          | Collation | Nullable |                    Default
-------------------+------------------------+-----------+----------+-----------------------------------------------
 product_key       | integer                |           | not null | nextval('dim_product_product_key_seq'::regclass)
 source_product_id | integer                |           | not null |
 product_name      | character varying(200) |           | not null |
 category          | character varying(100) |           | not null |
 unit_price        | numeric(10,2)          |           | not null |
Indexes:
    "dim_product_pkey" PRIMARY KEY, btree (product_key)
    "dim_product_source_product_id_key" UNIQUE CONSTRAINT, btree (source_product_id)
Referenced by:
    TABLE "fact_sales" CONSTRAINT "fact_sales_product_key_fkey" FOREIGN KEY (product_key) REFERENCES dim_product(product_key)
```

\d dim_employee

```
warehouse=# \d dim_employee
                                         Table "public.dim_employee"
       Column        |          Type          | Collation | Nullable |                     Default
---------------------+------------------------+-----------+----------+-----------------------------------------------
 employee_key        | integer                |           | not null | nextval('dim_employee_employee_key_seq'::regclass)
 source_employee_id  | integer                |           | not null |
 first_name          | character varying(100) |           | not null |
 last_name           | character varying(100) |           | not null |
 email               | character varying(255) |           | not null |
 job_title           | character varying(100) |           | not null |
 department_name     | character varying(100) |           | not null |
 department_location | character varying(100) |           | not null |
Indexes:
    "dim_employee_pkey" PRIMARY KEY, btree (employee_key)
    "dim_employee_source_employee_id_key" UNIQUE CONSTRAINT, btree (source_employee_id)
```

\d fact_sales

```
warehouse=# \d fact_sales
                                    Table "public.fact_sales"
     Column      |     Type      | Collation | Nullable |                   Default
-----------------+---------------+-----------+----------+---------------------------------------------
 sale_key        | integer       |           | not null | nextval('fact_sales_sale_key_seq'::regclass)
 date_key        | integer       |           | not null |
 customer_key    | integer       |           | not null |
 product_key     | integer       |           | not null |
 source_order_id | integer       |           | not null |
 quantity        | integer       |           | not null |
 unit_price      | numeric(10,2) |           | not null |
 total_price     | numeric(10,2) |           | not null |
Indexes:
    "fact_sales_pkey" PRIMARY KEY, btree (sale_key)
    "fact_sales_source_order_id_key" UNIQUE CONSTRAINT, btree (source_order_id)
    "idx_fact_sales_customer_key" btree (customer_key)
    "idx_fact_sales_date_key" btree (date_key)
    "idx_fact_sales_product_key" btree (product_key)
Foreign-key constraints:
    "fact_sales_customer_key_fkey" FOREIGN KEY (customer_key) REFERENCES dim_customer(customer_key)
    "fact_sales_date_key_fkey" FOREIGN KEY (date_key) REFERENCES dim_date(date_key)
    "fact_sales_product_key_fkey" FOREIGN KEY (product_key) REFERENCES dim_product(product_key)
```

Task 3.2: Compare Operational vs Warehouse Design

Unlike the HR and sales databases where information is split across multiple tables, the warehouse brings everything together. dim_customer and dim_product hold the same information that already exists in the sales database, this includes name, email, city, category, price but each row also includes a source_customer_id or source_product_id. These columns store the original ID from the sales database, so the warehouse can always trace each row back to where it came from.

The HR database needs a join every time you want an employee's department name or location, because that information sits in a completely separate table. dim_employee does away with that which department_name and department_location live right on the employee row. It feels like a step backward from what normalization teaches, but warehouses are not built for constant updates the way operational databases are. They are built for reading, and reading fast. Flattening the data out means one less join on every single query, which adds up quickly at scale.

dim_date is probably the most deliberate design choice in the whole schema. order_date in the sales database is just a date nothing more. Every time a query needs to know the month, the quarter, whether it was a weekend, that has to be calculated on the fly. dim_date pre-computes all of that and stores it as actual columns, so analysts can filter and group by time without writing any extra logic.

Task 3.3: Identify Facts vs Dimensions

Fact tables store measurable business events like sales transactions with numeric measures such as quantity and price, while dimension tables store the descriptive context around those facts like customer names, product categories, and dates. The fact_sales table is the fact table, and dim_customer, dim_product, dim_date, and dim_employee are the dimension tables. You can tell by looking at the columns, fact_sales stores numbers that can be measured and aggregated, like quantity, unit_price, and total_price. The dimension tables

store descriptive attributes like names, categories, and locations. The fact table also holds all the foreign keys that point to the dimension tables, which is the structural giveaway.

Each row in fact_sales represents one customer buying one product on one specific date that is the grain of the table. To answer something like 'which product category generated the most sales in the first week of February,' you would join fact_sales with dim_product and dim_date. fact_sales gives you the total_price to add up, dim_product gives you the category to group by, and dim_date lets you filter by the specific dates in that first week. None of that is possible without bringing all three tables together.

Part 4: ETL Pipeline

Task 4.1: Analyze the ETL Script

```
Container week03_hr_db Waiting
Container week03_warehouse_db Waiting
Container week03_hr_db Healthy
Container week03_sales_db Healthy
Container week03_warehouse_db Healthy
Container part01-etl-service-run-c9a78ef0d52d Creating
Container part01-etl-service-run-c9a78ef0d52d Created
============================================================
ETL Pipeline Starting
============================================================

[EXTRACT] Reading from operational databases...
  Extracted 8 customers from sales-db
  Extracted 10 products from sales-db
  Extracted 27 orders from sales-db
  Extracted 12 employees from hr-db

[TRANSFORM] Preparing dimension data...
  Generated 365 dim_date rows for 2026
  Employee dimension: denormalized with department info (done in extract query)

[LOAD] Writing to data warehouse...
  Loaded dim_date (365 rows, conflicts skipped)
  Loaded dim_customer (8 rows, upserted)
  Loaded dim_product (10 rows, upserted)
  Loaded dim_employee (12 rows, upserted)
  Loaded fact_sales (27 inserted, 0 already existed)

============================================================
ETL Pipeline Complete
============================================================
```

```
PS C:\Users\ksogodogo\ism6562s26-class\midterm\part01> docker compose run etl-service cat /etc/cron.d/etl-cron
time="2026-03-29T20:36:00-04:00" level=warning msg="Found orphan containers ([part01-etl-service-run-c9a78ef0d52d part01-etl-service-run
-3f0a11ce169a]) for this project. If you removed or renamed this service in your compose file, you can run this command with the --remov
e-orphans flag to clean it up."
[+]  3/3t 3/33
 ✔Container week03_warehouse_db Running                                                                                    0.0s
 ✔Container week03_hr_db        Running                                                                                    0.0s
 ✔Container week03_sales_db     Running                                                                                    0.0s
Container week03_hr_db Waiting
Container week03_warehouse_db Waiting
Container week03_sales_db Waiting
Container week03_hr_db Healthy
Container week03_sales_db Healthy
Container week03_warehouse_db Healthy
Container part01-etl-service-run-45f4816c6261 Creating
Container part01-etl-service-run-45f4816c6261 Created
# Run the ETL pipeline every 5 minutes, sending output to Docker logs
*/5 * * * * cd /app && /usr/local/bin/python etl_script.py >> /proc/1/fd/1 2>&1
```

Extract phase pulls data from both operational databases. From sales-db it reads customers, products, and orders. From hr-db it reads employees and departments. The script uses standard SELECT queries to pull all rows from each table. We know this because the output showed exact counts: 8 customers, 10 products, 27 orders and 12 employees extracted.

Transform phase handles two things. For dim_date, instead of pulling dates from any source database, the script generates them programmatically, it loops through every day in 2026 and calculates the year, quarter, month, month name, day of month, day of week, day name, and weekend flag for each date. For dim_employee, the denormalization happens directly in the extract query by joining employees and departments in hr-db before the data reaches the warehouse, so department_name and department_location land on the employee row.

Load phase uses ON CONFLICT clauses to ensure idempotency, meaning the script can run multiple times safely without creating duplicates. When a row already exists, identified by its source ID, the database either skips it or updates it in place rather than inserting a second copy. This is what the output meant by "conflicts skipped" and "upserted." The cron schedule */5 * * * * means the script runs every 5 minutes via cron.

Task 4.2: Verify ETL Results

Source database counts:

```
sales=# -- Run on sales-db
SELECT 'customers' AS source_table, COUNT(*) FROM customers
UNION ALL SELECT 'products', COUNT(*) FROM products
UNION ALL SELECT 'orders', COUNT(*) FROM orders;
 source_table | count
--------------+-------
 customers    |     8
 products     |    10
 orders       |    27
(3 rows)
```

Warehouse counts:

```
warehouse=# -- Run on warehouse-db
SELECT 'dim_customer' AS warehouse_table, COUNT(*) FROM dim_customer
UNION ALL SELECT 'dim_product', COUNT(*) FROM dim_product
UNION ALL SELECT 'dim_employee', COUNT(*) FROM dim_employee
UNION ALL SELECT 'dim_date', COUNT(*) FROM dim_date
UNION ALL SELECT 'fact_sales', COUNT(*) FROM fact_sales;
 warehouse_table | count
-----------------+-------
 dim_customer    |     8
 dim_product     |    10
 dim_employee    |    12
 dim_date        |   365
 fact_sales      |    27
(5 rows)
```

Task 4.3: Test the ETL Pipeline

ETL pipeline: New data from the operational databases.

Steps:

Insert a new order into the sales database:

```
sales=# INSERT INTO orders (customer_id, product_id, quantity, total_price, order_date)
VALUES (2, 1, 1, 1299.99, '2026-02-25');
INSERT 0 1
```

Verify the new order exists in the sales database:

```
sales=# SELECT * FROM orders ORDER BY order_id DESC LIMIT 3;
 order_id | customer_id | product_id | quantity | total_price | order_date
----------+-------------+------------+----------+-------------+------------
       28 |           2 |          1 |        1 |     1299.99 | 2026-02-25
       27 |           7 |         10 |        1 |      349.99 | 2026-02-22
       26 |           3 |          9 |        1 |      199.99 | 2026-02-20
(3 rows)
```

Manually trigger the ETL pipeline:

```
[+]  3/3t 3/33
 ✔Container week03_warehouse_db Running                                                                    0.0s
 ✔Container week03_hr_db        Running                                                                    0.0s
 ✔Container week03_sales_db     Running                                                                    0.0s
Container week03_warehouse_db Waiting
Container week03_sales_db Waiting
Container week03_hr_db Waiting
Container week03_sales_db Healthy
Container week03_warehouse_db Healthy
Container week03_hr_db Healthy
Container part01-etl-service-run-f78bcb2a64df Creating
Container part01-etl-service-run-f78bcb2a64df Created
============================================================
ETL Pipeline Starting
============================================================

[EXTRACT] Reading from operational databases...
  Extracted 8 customers from sales-db
  Extracted 10 products from sales-db
  Extracted 28 orders from sales-db
  Extracted 12 employees from hr-db

[TRANSFORM] Preparing dimension data...
  Generated 365 dim_date rows for 2026
  Employee dimension: denormalized with department info (done in extract query)

[LOAD] Writing to data warehouse...
  Loaded dim_date (365 rows, conflicts skipped)
  Loaded dim_customer (8 rows, upserted)
  Loaded dim_product (10 rows, upserted)
  Loaded dim_employee (12 rows, upserted)
  Loaded fact_sales (0 inserted, 28 already existed)

============================================================
ETL Pipeline Complete
============================================================
```

Verify the new order appears in the warehouse:

```
warehouse=# SELECT f.sale_key, d.full_date, c.first_name, c.last_name,
       p.product_name, f.quantity, f.total_price
FROM fact_sales f
JOIN dim_date d ON f.date_key = d.date_key
JOIN dim_customer c ON f.customer_key = c.customer_key
JOIN dim_product p ON f.product_key = p.product_key
ORDER BY f.sale_key DESC
LIMIT 3;
 sale_key | full_date  | first_name | last_name |     product_name     | quantity | total_price
----------+------------+------------+-----------+----------------------+----------+-------------
       55 | 2026-02-25 | Bob        | Smith     | Laptop Pro 15        |        1 |     1299.99
       27 | 2026-02-22 | Grace      | Wilson    | Monitor 27-inch      |        1 |      349.99
       26 | 2026-02-20 | Carol      | Williams  | Noise-Cancel Headset |        1 |      199.99
(3 rows)
```

Part 5: Analytical Queries

Task 5.1: Write Analytical Queries

Query 1: Monthly Sales Summary

```
SELECT
    d.month_name,
    d.year,
    COUNT(f.sale_key)        AS total_orders,
    SUM(f.total_price)       AS total_revenue
FROM fact_sales f
JOIN dim_date d ON f.date_key = d.date_key
GROUP BY d.year, d.month, d.month_name
ORDER BY d.year, d.month;
```

```
warehouse=# SELECT
    d.month_name,
    d.year,
    COUNT(f.sale_key)        AS total_orders,
    SUM(f.total_price)       AS total_revenue
FROM fact_sales f
JOIN dim_date d ON f.date_key = d.date_key
GROUP BY d.year, d.month, d.month_name
ORDER BY d.year, d.month;
 month_name | year | total_orders | total_revenue
------------+------+--------------+---------------
 January    | 2026 |           15 |       5773.78
 February   | 2026 |           13 |       5911.85
(2 rows)
```

Query 2: Top Customers by Spending

```
SELECT
    c.first_name,
    c.last_name,
    COUNT(f.sale_key)        AS number_of_orders,
    SUM(f.total_price)       AS total_spent
FROM fact_sales f
JOIN dim_customer c ON f.customer_key = c.customer_key
GROUP BY c.customer_key, c.first_name, c.last_name
ORDER BY total_spent DESC;
```

```
warehouse=# SELECT
    c.first_name,
    c.last_name,
    COUNT(f.sale_key)        AS number_of_orders,
    SUM(f.total_price)       AS total_spent
FROM fact_sales f
JOIN dim_customer c ON f.customer_key = c.customer_key
GROUP BY c.customer_key, c.first_name, c.last_name
ORDER BY total_spent DESC;
 first_name | last_name | number_of_orders | total_spent
------------+-----------+------------------+-------------
 Bob        | Smith     |                5 |     2391.94
 Alice      | Johnson   |                5 |     2209.94
 Eva        | Davis     |                3 |     1979.96
 Frank      | Miller    |                3 |     1949.97
 Carol      | Williams  |                4 |     1737.93
 David      | Brown     |                3 |      739.95
 Grace      | Wilson    |                3 |      549.96
 Henry      | Taylor    |                2 |      125.98
(8 rows)
```

Query 3: Revenue by Product Category and Month

```
SELECT
    p.category,
    d.month_name,
    d.year,
    SUM(f.total_price)       AS total_revenue
FROM fact_sales f
JOIN dim_product p ON f.product_key = p.product_key
JOIN dim_date d    ON f.date_key    = d.date_key
GROUP BY p.category, d.month, d.month_name, d.year,
ORDER BY p.category, d.year, d.month;
```

```
warehouse=# SELECT
    p.category,
    d.month_name,
    d.year,
    SUM(f.total_price)       AS total_revenue
FROM fact_sales f
JOIN dim_product p ON f.product_key = p.product_key
JOIN dim_date d    ON f.date_key    = d.date_key
GROUP BY p.category, d.year, d.month, d.month_name
ORDER BY p.category, d.year, d.month;
  category   | month_name | year | total_revenue
-------------+------------+------+---------------
 Books       | January    | 2026 |        263.94
 Books       | February   | 2026 |        131.97
 Electronics | January    | 2026 |       3699.89
 Electronics | February   | 2026 |       4799.91
 Furniture   | January    | 2026 |       1809.95
 Furniture   | February   | 2026 |        979.97
(6 rows)
```

Task 5.2: EXPLAIN ANALYZE Comparison

On the sales (operational) database:

```
sales=# EXPLAIN ANALYZE
SELECT p.category,
       COUNT(*) AS order_count,
       SUM(o.total_price) AS total_revenue
FROM orders o
JOIN products p ON o.product_id = p.product_id
GROUP BY p.category
ORDER BY total_revenue DESC;
                                                        QUERY PLAN
-------------------------------------------------------------------------------------------------------------------------------
 Sort  (cost=53.21..53.48 rows=110 width=258) (actual time=0.090..0.092 rows=3 loops=1)
   Sort Key: (sum(o.total_price)) DESC
   Sort Method: quicksort  Memory: 25kB
   ->  HashAggregate  (cost=48.10..49.48 rows=110 width=258) (actual time=0.045..0.047 rows=3 loops=1)
         Group Key: p.category
         Batches: 1  Memory Usage: 24kB
         ->  Hash Join  (cost=12.47..38.58 rows=1270 width=234) (actual time=0.027..0.032 rows=28 loops=1)
               Hash Cond: (o.product_id = p.product_id)
               ->  Seq Scan on orders o  (cost=0.00..22.70 rows=1270 width=20) (actual time=0.005..0.006 rows=28 loops=1)
               ->  Hash  (cost=11.10..11.10 rows=110 width=222) (actual time=0.009..0.010 rows=10 loops=1)
                     Buckets: 1024  Batches: 1  Memory Usage: 9kB
                     ->  Seq Scan on products p  (cost=0.00..11.10 rows=110 width=222) (actual time=0.004..0.005 rows=10 loops=1)
 Planning Time: 0.787 ms
 Execution Time: 0.178 ms
(14 rows)
```

On the warehouse database:

```
warehouse=# EXPLAIN ANALYZE
SELECT p.category,
       COUNT(*) AS order_count,
       SUM(f.total_price) AS total_revenue
FROM fact_sales f
JOIN dim_product p ON f.product_key = p.product_key
GROUP BY p.category
ORDER BY total_revenue DESC;
                                                          QUERY PLAN
-----------------------------------------------------------------------------------------------------------------------------------
 Sort  (cost=47.16..47.43 rows=110 width=258) (actual time=0.058..0.059 rows=3 loops=1)
   Sort Key: (sum(f.total_price)) DESC
   Sort Method: quicksort  Memory: 25kB
   ->  HashAggregate  (cost=42.05..43.43 rows=110 width=258) (actual time=0.051..0.053 rows=3 loops=1)
         Group Key: p.category
         Batches: 1  Memory Usage: 24kB
         ->  Hash Join  (cost=12.47..34.78 rows=970 width=234) (actual time=0.032..0.037 rows=28 loops=1)
               Hash Cond: (f.product_key = p.product_key)
               ->  Seq Scan on fact_sales f  (cost=0.00..19.70 rows=970 width=20) (actual time=0.014..0.015 rows=28 loops=1)
               ->  Hash  (cost=11.10..11.10 rows=110 width=222) (actual time=0.011..0.011 rows=10 loops=1)
                     Buckets: 1024  Batches: 1  Memory Usage: 9kB
                     ->  Seq Scan on dim_product p  (cost=0.00..11.10 rows=110 width=222) (actual time=0.006..0.007 rows=10 loops=1)
 Planning Time: 0.215 ms
 Execution Time: 0.096 ms
(14 rows)
```

Both queries follow the same execution path. a Hash Join between the sales/orders table and the product table, followed by a HashAggregate to group by category, and a final sort. The scan types are the same in both cases: sequential scans on every table. Neither query uses index scans, which makes sense because with only 28 rows and 10 products, the database decides it's faster to just read everything directly rather than look up an index.

The most noticeable difference is in execution time. The warehouse query completed in 0.096ms while the sales database query took 0.178ms, nearly twice as long. The planning time also differed: the warehouse took 0.215ms to plan versus 0.787ms for the sales database. This suggests the warehouse schema is slightly simpler for the query planner to reason about, likely because the star schema has a more predictable structure.

With millions of rows these differences would become much more significant. The warehouse is designed specifically for this type of analytical query which wide reads across large datasets grouped by category. The star schema means fewer joins are needed to get to useful attributes. In the sales database, adding more dimensions like date or region would require additional joins to normalized tables. In the warehouse those attributes are already pre-joined into dimension tables, making complex analytical queries faster and easier to execute at scale.